Quadrupole mass filtering in MScan

Rod voltages, scan law and parameters

ESIBD Explorer Plugins, v0.5.10

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1 Scope

A scan sweeps one quantity and, optionally, the DC of the two rod pairs through two AMPR channels. The ordinary line sweeps the RF amplitude A of one AMX output (CH0-CH1 or CH2-CH3). A map line instead sweeps the axis potential x, the entry stub, or the AMX frequency at fixed A (Section 2). At each point MScan averages the current of every enabled DMMR module, the ion collector among them, and saves current versus the swept quantity for each. Apart from the frequency line, which sets the AMX oscillator through the AMX HD plugin, a scan does not change the AMX frequency, duty cycle or routing. The file keeps the swept quantity as the axis; on an amplitude line m/z is derived from it with the settings of the scan (Section 5) and saved apart, as `Validation/mz`.

2 Scan axes

A line sweeps exactly one quantity; every other setting it needs is applied before it starts and confirmed (the AMX frequency to 10^{-6} relative, the channels to Voltage tolerance).

- **Amplitude line** — the ordinary scan, and every map line that measures a peak: the RF amplitude A of one AMX output, with the DC law of equation (3). It is the only axis from which m/z is computed.
- **Axis line** (`offset`): the common potential x at fixed A, so U and the rod RF stay constant while only the ion energy changes.
- **Entry-stub line** (`extra`): the rod pairs at fixed A and x, then the Stub_Entry_Q2 AMPR channel over its own span.
- **Frequency line** (`frequency`): the AMX oscillator at fixed A, set through the AMX HD plugin, which keeps the 50 % duty cycle. Since $m/z \propto 1/f^2$, it moves the transmitted window without touching U/A.

The file keeps the swept quantity as its axis. On an axis or entry-stub line A and f do not change, so m/z is the same at every point; on a frequency line it varies as $1/f^2$. Neither axis is A, so `Validation/mz` is written for amplitude lines only: elsewhere the axis in the file, the recorded f and `Validation/setup` carry the same information.

3 Rod potentials

The PSU rails of the selected AMX output are set to $V_{\mathrm{pos}} = +A$ and $V_{\mathrm{neg}} = -A$; the AMX switches each rod pair between them in opposite phase, so with $s(t) = \pm 1$ a square wave of 50 % duty at the AMX frequency $f = \Omega/2\pi$:

$$\Phi_1(t) = x + U + A\,s(t) \qquad \text{pair 1 (AMPR channel Rod pair 1)} \tag{1}$$

$$\Phi_2(t) = x - U - A\,s(t) \qquad \text{pair 2 (AMPR channel Rod pair 2)} \tag{2}$$

These are equations (1) and (2) of Steel and Henchman [1] ($V_{2,4} = +U + V_0 \cos \omega t$, $V_{1,3} = -U - V_0 \cos \omega t$) with $V_0 = A$, the cosine replaced by $s(t)$, and a common potential x (the axis). Adding x to all four rods does not change the filter: it sets the potential of the axis, hence the axial energy of the ions. Which pair receives $+U$ does not matter either: with 50 % duty, swapping the pairs only shifts the wave by half a period. Figure 1 shows an example.

Assumptions not yet verified on the instrument. (i) Each rod swings between $+A$ and $-A$ around its DC, i.e. $V_0 = A$ (the AMX switches between the PSU rails, referenced to the DC applied to that pair). (ii) The AMPR DC reaches the rods of the pair driven by the same AMX connector. (iii) The wave on the rods stays square. A scope on the rods (or on the AMX *Monitor* outputs, 1:100, if the AMX HD has them as the 4ED does) settles all three. The lab scan of 2026-11-02 14:34 supports (ii): the collector current responds to the difference between the two DC values and peaks when they are equal.

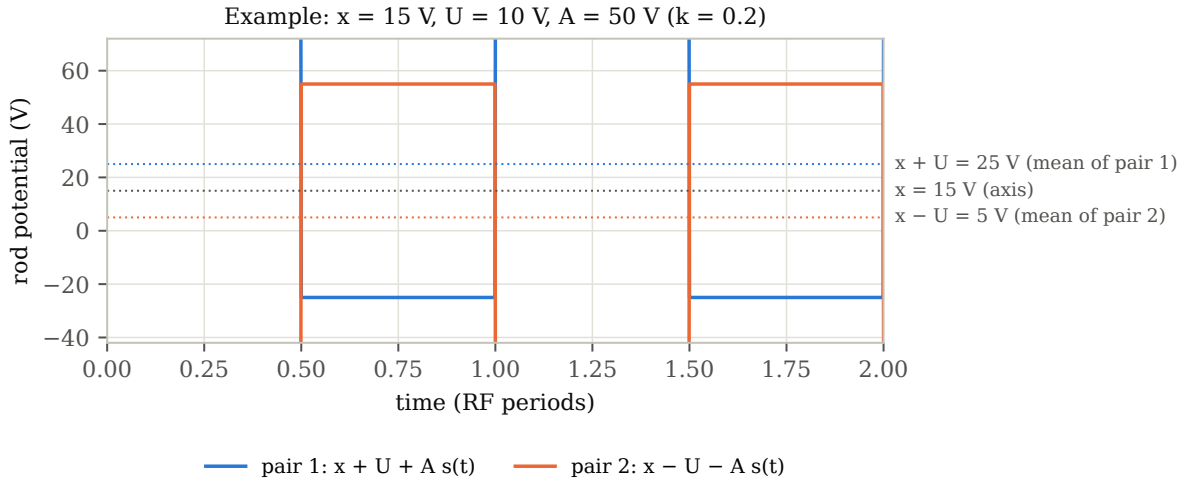


Figure 1: Rod potentials at one scan point. Only the antisymmetric part ($\pm U$, $\pm A\,s$) filters masses; x is common to both pairs.

4 DC law and parameters

For a coefficient $k > 0$, MScan commands at every amplitude

$$U(A) = \max\big(U_0,\ k\,A - \delta\big), \qquad \Phi^{\mathrm{DC}}_{1,2} = x \pm U(A). \tag{3}$$

$k = 0$ is RF-only: both pairs stay at x, without the floor U_0. A negative k swaps the pairs (U is computed with $|k|$). With $\delta = 0$ and $k\,A \geq U_0$, equation (3) is the scan line of the textbook [1]: $U/A = k$ constant, a straight line through the origin.

Symbol	MScan setting	Unit	Default	Role
A	Amplitude from / to / step	V	–	scanned RF amplitude, per rod (rails $\pm A$)
x	Axis potential x	V	0	common potential of the four rods, i.e. the axis; sets the ion energy, does not filter. Keep the value the rods hold outside a scan.
k	U/V ratio k	–	0.2	slope U/A of the scan line; sets the resolution. 0 = RF-only, negative swaps the pairs.
δ	Resolution δ	V	0	lowers the line: $U/A = k - \delta/A$ rises with A, so peaks keep about the same width in A (Section 5). Must be ≥ 0.
U_0	Minimum U	V	0.5	floor of U, must be > 0 (Section 5).
–	Rod pair 1 / Rod pair 2	–	None	AMPR channels of the two pairs. Pair 2 alone is refused.

Table 1: Parameters of the DC law, in the Mass filter group of the MScan panel, which also shows a Preview of U and of both pair potentials over the scan. Settings files and HDF5 scans keep the earlier keys: Fixed offset (x), Offset coefficient (k), Resolution offset (δ), Quadrupole offset and Quadrupole offset pair 2.

Do not confuse k with the α of Steel and Henchman. Their α is the Mathieu parameter $a = 8eU/(mr_0^2\Omega^2)$; the ratio they keep constant is $U/V_0 = \alpha/(2q) = 0.1678$ for a cosine. Here $k = a/(2q)$, and its useful values are near 0.2 because the wave is square (Section 5). Entering 0.233 as k would put the line above the apex: nothing passes.

One pair only. If only pair 1 is selected, pair 2 keeps its own value V_2. The filtering DC is then $(x + U - V_2)/2$ and the axis moves to $(x + U + V_2)/2$ during the scan. Use both pairs for equation (3).

At each point. The rails are commanded first, then both rod pairs (rounded to the AMPR display precision). No current is acquired until the AMPR confirms each target by its setpoint readback and its Monitor is within Voltage tolerance. Start is refused if a value is not finite, $\delta < 0$, $U_0 \leq 0$, a target over the span leaves the channel Min/Max or AMPR rating, or a channel is OFF or under equation control. Normal completion restores the initial pair potentials with the PSU setpoints; Stop or an error holds the last values.

Recorded. The setup keeps `offset = {coefficient, fixed_v, resolution_offset_v, minimum_u_v, pairs}` (channel, AMPR module and channel, sign, initial value)}; every point keeps the commanded pair potentials `offset_target` and their Monitors `offset_v`, one column per pair, next to `rail_v`, `rail_i`, sample counts and point status.

5 What the parameters do

With the conventions of [1] (U and $V_0 = A$ per rod, from the axis),

$$a = \frac{8eU}{mr_0^2\Omega^2}, \qquad q = \frac{4eA}{mr_0^2\Omega^2}, \qquad \frac{a}{q} = 2\,\frac{U}{A} = 2k \quad (\delta = 0). \tag{4}$$

All ions lie on the line $a = 2kq$; an ion passes when its point is inside the stability region (Figure 2). For a 50 % square wave the apex is at $a = 0.2381$, $q = 0.5529$, i.e. $k = 0.2153$ (cosine: 0.2370, 0.7060, $k = 0.1678$). The RF-only limit is $q = 0.712$ (cosine 0.908).

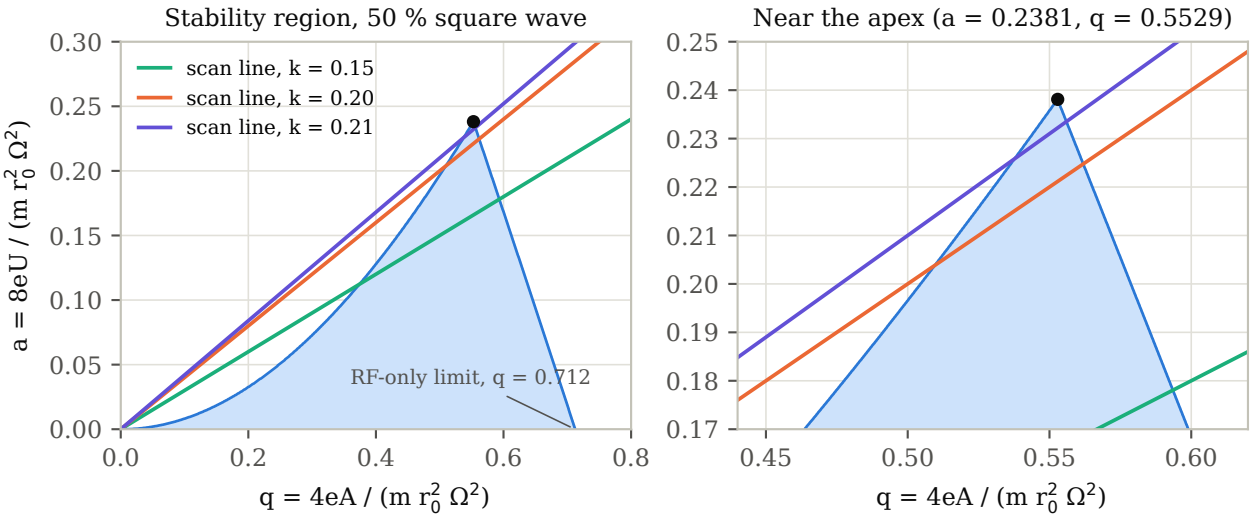


Figure 2: Stability region of the 50 % square wave (both directions stable) and scan lines $a = 2kq$. Computed with the transfer matrix of the piecewise-constant Hill equation.

k	transmitted q band	centre $\bar{q}$	$R = \bar{q}/\Delta q$
0	0 – 0.712	–	high-pass
0.10	0.246 – 0.629	0.438	1.1
0.15	0.375 – 0.593	0.484	2.2
0.18	0.455 – 0.574	0.515	4.3
0.19	0.482 – 0.568	0.525	6.1
0.20	0.510 – 0.562	0.536	10.3
0.205	0.524 – 0.559	0.542	15.4
0.21	0.538 – 0.556	0.547	30.4
0.2125	0.545 – 0.554	0.550	57

Table 2: Transmitted q band versus k ($\delta = 0$, ideal 50 % square wave). Near the apex $R \approx 0.157/(0.2153 - k)$. R is $m/\Delta m$ of the band at fixed A.

Peak position. An ion transmitted at the centre $\bar{q}$ of the band has

$$\frac{m}{z}\,[\mathrm{Th}] \approx 2.444\,\frac{A\,[\mathrm{V}]}{\bar{q}}\left(\frac{4\,\mathrm{mm}}{r_0}\right)^2\left(\frac{500\,\mathrm{kHz}}{f}\right)^2, \tag{5}$$

about $4.56\,A$ at $k = 0.2$ for $r_0 = 4$ mm and 500 kHz. On an amplitude line MScan computes it at every point, with $\bar{q}$ from U/V_0 there, the AMX frequency, r0 and the RF attenuation factor $c = V_0/A$, shows it as m/z range and on the plot (Plot axis), and saves it as `Validation/mz`. On 2026-11-03 the Agilent G2421A tuning mix gave its ions 622, 922, 1522 and 2122 on Q2 at 400, 450 and 500 kHz: $c = 0.978$ (band widths at $k = 0.20$, transmission at $k = 0.21$) and an effective $r_0 = 2.625$ mm, now the defaults: 5 % above the nominal 2.5 mm of Q2 (5 mm between opposite rods), about 10 % in m/z. The filter then closes at $k = 0.2153\,c = 0.2105$. In RF-only mode ($k = 0$) ions with $m/z < 3.43\,A$ (same units) are rejected, so a spectrum of several species is a staircase: each species removes a step at its cut-off amplitude A_c, and its peak at $k = 0.2$ lies near $0.753\,A_c$. The mass window scales as A/f^2: at fixed $A_{\max}$, 300 kHz reaches 2.8 times and 200 kHz 6.3 times the m/z of 500 kHz. The rail current also scales with f (Section 6).

Resolution and the amplitude on the rods. Table 2 assumes $V_0 = A$. If the rods see cA, the filter works at k/c: at $k = 0.20$ it closes completely for $c < 0.929$, at $k = 0.21$ for $c < 0.975$. The scan itself measures c: raise k until the signal

disappears at k_c; then $c = k_c/0.2153$. Rounded edges matter much less: a trapezoid with rise times of 100, 200, 300 and 500 ns at 500 kHz moves the apex to $k = 0.2141$, 0.2110, 0.2063 and 0.1922; an exponential edge with $\tau = 50$, 100, 150 ns to 0.2123, 0.2046, 0.1937. With DC coupling, a duty error of 1 % moves the apex by about ± 0.018 in k.

Resolution δ. With $\delta = 0$ the peak width grows in proportion to A (constant R). With k at the apex value and $U = kA - \delta$, the width is nearly constant, $\Delta A \approx 6.4\,\delta$ (simulated: 6.8, 6.5, 6.4, 6.4 V at $A = 50$, 100, 200, 300 V for $\delta = 1$ V). Commercial instruments set slope and intercept on a light and a heavy reference ion, because the ideal constant-width line is a curve [2]. A negative δ (a positive constant added to kA) at the apex slope passes nothing, which is why MScan refuses $\delta < 0$.

Minimum U. At zero DC every ion heavier than the RF-only cut-off passes. Without a floor, a scan starting near $A = 0$ begins with an unfiltered plateau (simulated for an ion with its peak at 300 V and $\delta = 1$ V: transmitted from 0 to 4.72 V with $\max(0, \cdot)$, only from 4.57 to 4.72 V without clamping). Commercial residual gas analysers keep the DC from reaching zero ("Zero Blast Suppression" [3]). Peaks below $A^* = (U_0 + \delta)/k$ are not filtered reliably.

6 Limits of the present set-up

- Rail current of the Q2 output (AMX HD, 500 kHz, lab 2026-11-02): about 12 mA + 0.565 mA/V × A per rail, i.e. 153 mA at 250 V. With $I_{\mathrm{lim}} = 200$ mA the amplitude cannot exceed about 333 V at 500 kHz, even with the PSU range switched on. The current scales with f: about 97 mA at 300 kHz and 250 V.
- PSU headroom (PSU_A in Half range, lab 2026-11-03): output + dropout fell from 256 V at 0 mA to about 248.5 V at 150 mA, so the rails could not regulate above about 248 V and a 50–300 V scan stopped at 250 V. MScan now refuses a scan above output + dropout − 2 V, stops at once when a rail has lost its headroom, and flags points measured with less than 2 V (`low_headroom`). Full range raises the internal supply; check the dropout and the PSU temperature at the highest amplitude.
- The scan law is applied by two instruments (PSU and AMPR) that must agree to better than 1 % in ratio near the apex. Rectangular-wave (digital) quadrupoles can avoid that by keeping the amplitude and the DC fixed and sweeping the frequency: U/A is then constant by construction and $m/z \propto 1/f^2$. A frequency line does this (Section 2), but only when the AMX HD runs on its 100 MHz clock and its frequency can be set from MScan; the ordinary amplitude line, on which every number in this document was measured, keeps f fixed.
- A filter without DC is also possible with DC-coupled square waves by changing the duty cycle (computed: $R \approx 7$ at 60.0 %, 12 at 60.5 %, 41 at 61.0 %; closed above about 61.2 %). MScan does not do this: it leaves the AMX timing, and so the duty cycle, unchanged.

7 Recommended order on a new set-up

1. Confirm ions on the collector at a fixed amplitude: the current must be positive (for positive ions) and fluctuate with the spray. The electrospray current alone does not prove that ions enter.
2. Record one scan with the electrospray HV off: it gives the background of the collector, which MScan does not subtract.
3. Scan with $k = 0$ to the highest available A: count the cut-off steps and note their A_c.
4. Scan with $k = 0.15$, 0.18, 0.20, 0.21 ($\delta = 0$, $U_0 = 0.5$ V); the value where the signal vanishes gives c (amplitude on the rods / A).
5. Only then use δ, set on two reference ions, if constant peak width is needed.

References

[1] C. Steel, M. Henchman, *Understanding the Quadrupole Mass Filter through Computer Simulation*, J. Chem. Educ. 75 (1998) 1049.

[2] R. E. Pedder, *Practical Quadrupole Theory: Graphical Theory*, Extrel Application Note RA_2010A (ASMS 2001).

[3] Stanford Research Systems, *RGA120/220/320 Residual Gas Analyzer, User Manual*, rev. 1.08 (2025), section "Zero Blast Suppression".